Joint Commission Accreditation Readiness & Patient Safety Analytics Using SAS

Project Overview

This project applied SAS-based statistical modeling to evaluate simulated hospital patient safety measures designed around concepts commonly associated with Joint Commission accreditation and healthcare quality monitoring. The dataset was simulated for educational and portfolio purposes.

Objectives

- Simulate hospital patient safety indicators
- Build an accreditation readiness score
- Identify hospitals at elevated accreditation risk
- Apply logistic regression methods
- Rank hospitals using predicted risk probabilities

Methodology

A simulated dataset containing 300 hospitals was generated in SAS across Community, Teaching, Critical Access, and Psychiatric hospitals. Procedures used included DATA step programming, PROC LOGISTIC, PROC SGPLOT, PROC CORR, and PROC SORT.

Key Findings

Falls with injury increased accreditation risk (OR=1.88, $p<0.001$). Restraint utilization increased accreditation risk (OR=1.20, $p=0.001$). Higher medication reconciliation compliance reduced accreditation risk (OR=0.934, $p=0.018$).

Model Performance

C-statistic: 0.749
Percent concordant: 74.9%

Conclusion

Patient safety measures demonstrated stronger associations with accreditation risk than hospital classification. This project demonstrates SAS programming, healthcare quality analytics, logistic regression modeling, and interpretation of Joint Commission-inspired quality metrics.